

AT30K Configuration Review & Recommendations

✅ Correct Settings (No Changes Needed)

Basic Commands

php

```
"ADCommand" => "AD", // ✅ Correct
"APCommand" => "AP", // ✅ Correct
"ARCommand" => "AR", // ✅ Correct
"ADCommandClose" => "AD,FF", // ✅ Correct
"DRCommand" => "DR", // ✅ Correct
"ESCCommand" => "1B", // ✅ Correct
"PWCommand" => "PW", // ✅ Correct
"PWpassword" => "3179636424", // ✅ Correct
"JFCommand" => "JF", // ✅ Correct
"WDCommand" => "WD", // ✅ Correct
"CSCommnad" => "CS", // ✅ Correct (note typo but functional)
```

Memory Addresses

php

```
"CableData" => "6000,61ff", // ✅ Correct
"sensitivityMemoryAddress" => "6006", // ✅ Correct
"mapMemoryAddress" => "600c", // ✅ Correct
"DPCurrentMap" => "4040", // ✅ Correct
"SubnetMask" => "2104,2107", // ✅ Correct
"Gateway" => "2108,210b", // ✅ Correct
```

⚠️ Issues Found & Recommended Changes

1. Buffer Length Issues

php

```
// Current (may be insufficient)
"ARCommandBufferLength" => "110",

// Recommended (based on 4 archives × 32 chars + overhead)
"ARCommandBufferLength" => "150",
```

2. Memory Address Typo

```
php

// Current (extra space)
"criteriaTableMemoryAddress" => "600b ",

// Fixed
"criteriaTableMemoryAddress" => "600b",
```

3. Command Typo

```
php

// Current (typo in key name)
"CSCommnad" => "CS",

// Should be
"CSCommand" => "CS",
```

4. Map Data Inconsistency

```
php

// Current - Map5Data points to Map1Data address
"Map5Data" => "6219,621B",

// Should be (if you have 5 maps)
"Map5Data" => "7A05,7A07",
```

5. JR Command Format

```
php

// Current (specific to queue clearing)
"JRCommand" => "jr,05e000 00 00 00 00",

// Should be generic
"JRCommand" => "JR",
"QueueClearCommand" => "jr,05e000 00 00 00 00",
```

Additional Recommended Settings

Connection Timeouts

php

```
"connection_timeout" => 10,           // seconds
"command_timeout" => 30,              // seconds
"data_retrieval_timeout" => 60,       // seconds
```

Device Type Detection

php

```
"device_type_detection" => true,
"at30k_identifier" => "MON 30A>",
"at20k_identifier" => "MON 24M>",
```

Error Handling

php

```
"max_retry_attempts" => 3,
"retry_delay" => 2,           // seconds
"error_log_enabled" => true,
```

Protocol Validation

php

```
"validate_responses" => true,
"expected_prompt" => "MON 30A>",
"validate_checksums" => true,
```



Complete Updated Configuration

php

```
<?php
return [
    // Existing settings (keep as-is)...

    // Fixed issues
    "ARCommandBufferLength" => "150",
    "criteriaTableMemoryAddress" => "600b",
    "CSCCommand" => "CS", // Fixed typo
    "Map5Data" => "7A05,7A07", // If using 5 maps

    // Split JR command
    "JRCommand" => "JR",
    "QueueClearCommand" => "jr,05e000 00 00 00 00",

    // Additional recommended settings
    "connection_timeout" => 10,
    "command_timeout" => 30,
    "data_retrieval_timeout" => 60,
    "max_retry_attempts" => 3,
    "retry_delay" => 2,
    "validate_responses" => true,
    "expected_prompt" => "MON 30A>",
    "device_type_detection" => true,
    "at30k_identifier" => "MON 30A>",

    // Keep all other existing settings...
];
```

Testing Your Configuration

1. Basic Connectivity Test

php

```
// Test sequence
1. ESC command
2. AD,01 (or your system ID)
3. PW,3179636424 (if needed)
4. CS (cable status)
5. AD,FF (close)
```

2. Data Retrieval Test

php

// Test data commands

1. DR,20FD,20FD (get system ID)
2. DP,1,1,6000,61FF (get cable 1 data)
3. AP (get archive pointer)

3. Validation Points

- Check buffer lengths accommodate actual responses
- Verify memory addresses return expected data
- Test timeout values under various network conditions
- Confirm error handling works properly

Critical Notes

Buffer Sizing

- Monitor actual response lengths during testing
- Increase buffer sizes if you get truncated responses
- Different AT30K firmware versions may have varying response lengths

Network Considerations

- Default Telnet port is 23
- Some AT30K devices use different ports (502, 1001, 1024)
- Ensure your `service_port` setting matches device configuration

Security

- The password `3179636424` is the default restricted access password
- Consider implementing session management to prevent conflicts
- Always use `AD,FF` to properly close connections

Performance

- The 1000ms pagination setting may be too aggressive for slow networks
- Consider increasing timeouts for unreliable connections
- Implement proper error recovery mechanisms

